12MT101CV

# MATHEMATICS-I

(Common to all Branches)

| | | | |
|---|---|---|---|
| Instruction | : 5 periods/week | Sessional marks | : 25 |
| Tutorials | : 1 period/week | End Examination | : 75 Marks |
| Credits | : 5 | Duration | : 3 Hours |

**UNIT – I**

**Differential Equations of First Order and their applications:** Differential equations of first order and first degree – exact, linear and Bernoulli. Applications to Newton's Law of cooling, Law of natural growth and decay, orthogonal trajectories.

**UNIT – II**

**Higher Order Linear Differential Equations and their applications:** Non-homogeneous linear differential equations of second and higher order with constant coefficients with RHS term of the type $e^{ax}$, Sinax, Cosax, Polynomials in x, $e^{ax}V(x)$, $x^nV(x)$, Method of variation of parameters. Applications to bending of beams, electrical circuits, simple harmonic motion.

**UNIT – III**

**Functions of single and several variables:** Rolle's Theorem – Lagrange's Mean Value Theorem – Cauchy's mean value Theorem – Generalized Mean Value theorem (all theorems without proof) Functions of several variables – Functional dependence- Jacobian- Maxima and Minima of functions of two variables with constraints and without constraints

**UNIT – IV**

**Applications of single variable:** Radius, Centre and Circle of Curvature – Evolutes and Envelopes



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**UNIT – V**

**Laplace Transforms:** Laplace transform of standard functions – Inverse transform – first shifting Theorem, Transforms of derivatives and integrals – Unit step function – second shifting theorem – Dirac's delta function - Convolution theorem – Periodic function - Differentiation and integration of transforms-Application of Laplace transforms to ordinary differential equations.

**UNIT – VI**

**Multiple integrals:** Double and Triple integrals – change of variables – change of order of integration.

**UNIT – VII**

**Vector Differentiation:** Gradient- Divergence- Curl and their related properties of sums- products- Laplacian and second order operators.

**UNIT – VIII**

**Vector Integration:** Vector Integration - Line integral – work done – Potential function – area- surface and volume integrals Vector integral theorems: Green's theorem-Stoke's and Gauss's Divergence Theorem (With out proof). Verification of Green's - Stoke's and Gauss's Theorems.

**TEXT BOOKS**

1. A text Book of Engineering Mathematics, Vol-1,T. K. V. Iyengar, B. Krishna Gandhi and Others, S. Chand & Company.
2. Engineeering Mathematics, Vol-1, Dr.D.S. Chandra sekharaiah, PRISM Publications.

**REFERENCES**

1. Engineering Mathematics-1, Dandala Radhakrishna Reddy, R.L.naidu, Ch.Purnachandra rao, Tata McGraw Hill.
2. Advanced Engineering Mathematics, Ervin Kreyszig, Wiley India Pvt. Ltd.
3. A text Book of Engineering Mathematics, Shahnaz Bathul, Right Publishers.



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12MT151CV

# MATHEMATICS-II

(Common to all branches)

| | | | |
|---|---|---|---|
| Instruction | : 5 periods/week | Sessional marks | : 25 |
| Tutorials | : 1 period/week | End Examination | : 75 Marks |
| Credits | : 5 | Duration | : 3 Hours |

**UNIT – I**

**Solution for linear systems:** Matrices and Linear system of equations: Elementary row transformations-Rank-Echelon form, Normal form – Solution of Linear Systems: consistency and inconsistency of system of equations.

**UNIT – II**

**Eigen values and Eigen vectors:** Eigen values, Eigen vectors – properties – Condition number, Cayley-Hamilton Theorem (with proof) - Inverse and powers of a matrix by Cayley-Hamilton theorem – Diagonalization of matrix. Calculation of powers of matrix – Modal and spectral matrices.

**UNIT – III**

**Linear Transformations and Quadratic Forms:** Real matrices – Symmetric, skew - symmetric, orthogonal, Linear Transformation – Orthogonal Transformation. Complex matrices: Hermitian, Skew-Hermitian and Unitary – Eigen values and eigen vectors of complex matrices and their properties. Quadratic forms: Reduction of quadratic form to canonical form – Rank - nature - index - signature - Sylvester law (without proof).

**UNIT – IV**

**Solution of Non linear Equation and Curve Fitting:** Solution of Algebraic and Transcendental Equations: Introduction – The bisection Method – The Method of False Position – The Iteration Method – Newton-Raphson Method. Curve fitting: Fitting a straight line –Second degree curve-exponential curve-power curve by method of least squares.



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**UNIT – V**

**Interpolation and Numerical differentiation:** Introduction-Errors in Polynomial Interpolation – Finite differences- Forward differences- Backward differences –Central differences – Symbolic relations and separation of symbols-Differences of a polynomial-Newton's formulae for interpolation – Gauss Central difference formulae –Interpolation with unevenly spaced points-Lagrange's Interpolation formula – Numerical Differentiation.

**UNIT – VI**

**Numerical Solution of Initial Value Problems in ODE:** Solution by Taylor's series-Picard's Method of successive approximations-Euler's Method-Runge-Kutta Methods.

**UNIT – VII**

**Fourier series:** Determination of Fourier coefficients – Fourier series – even and odd functions – Fourier series in an arbitrary interval – even and odd periodic continuation – Half-range Fourier sine and cosine expansions.

**UNIT – VIII**

**Partial differential Equations:** Introduction and Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions –solutions of first order linear (Lagrange) equation and nonlinear (standard type) equations, Method of separation of variables.

**TEXTBOOKS**

1. Mathematical Methods, T. K. V. Iyengar, B. Krishna Gandhi and Others, S. Chand & Company.
2. A text book of Mathematical Methods, Shahnaz Bathul, Right Publisshers.
3. Mathematical Methods, Dr.D.S.C.PRISM Publications.



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12MT202CV

With effect from academic year 2013-2014

# MATHEMATICS-III

| | | | |
|---|---|---|---|
| Instruction | : 3 Periods/Week | Sessional Marks | : 25 |
| Tutorial | : 1 Period/Week | End Examination | : 75 Marks |
| Credits | : 3 | Duration | : 3 Hours |

**UNIT-I**

**SPECIAL FUNCTIONS I**: Gamma and Beta functions – properties- Evaluation of improper integrals, Series solutions to differential equations, Bessel functions –properties – Recurrence relations – Orthogonality.

**UNIT –II**

**SPECIAL FUNCTIONS II**:Legendre polynomials—Properties – Rodrigue's formula- recurrence relations – Orthogonality.

**UNIT-III**

**FUNCTIONS OF COMPLEX VARIABLE:** Continuity-Differentiability-Analyticity-Properties-Cauchy-Riemann equations in Cartesian and polar coordinates, Harmonic and conjugate harmonic functions –Milne's- Thompson method.

**UNIT-IV**

**ELEMENTARY FUNCTIONS:**Exponential, trigonometric, hyperbolic functions and their properties- General power $Z^c$ (c is complex) principal value, Logarithmic function.

**UNIT-V**

**COMPLEX INTEGRATION:** Line integral-Evaluation along a path and by indefinite integration- Cauchy's integral theorem-Cauchy's integral formula- Generalized integral formula.

**UNIT-VI**

**COMPLEX POWER SERIES:** Radius of convergence-Expansion in Taylor's series, Maclaurin's series and Laurent's series. Singular point –Isolated singular point-pole of/order m-Essential singularity.



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**UNIT-VII**

**CONTOUR INTEGRATION:** Residue-Evaluation of residue by formula and by Laurent series-Residue Theorem. Evaluation of integrals of the type:

A: Improper real integral $\int_{-\infty}^{\infty} f(x)dx$

B: $\int_{\alpha}^{\alpha+2\pi} f(\cos u, \sin u)\, du$

C: Fourier integrals $\int_{-\infty}^{\infty} e^{imx} f(x)\, dx$

D; Integrals by indentation.

**UNIT-VIII**

**CONFORMAL MAPPING:** Transformation by $e^z$, $\ln z$, $z^2$, $z^n$ (n a positive integer) sinz, cosz z+a/z. Translation, rotation, inversion and bilinear transformation. Fixed point – Cross ratio – Properties- Invariance of circles and cross ratio-determination of bilinear transformation mapping 3 given points.

**TEXT BOOKS :**

1. Engineering Mathematics-III, T.K.V.Iyyengar, B.Krishna Ghandhi & others, 9th Edition, S.Chand,2011
2. Engineering Mathematics-III, P.B.Bhaskara Rao, S.K.V.S. Rama Chary,S.Bhujanga Rao, 2nd Edition, BSP,2010

**RFERENCES :**

1. Special Functions & Complex Variables (Engineering Mathematics-III), Shahanaz Bathul, 2nd Edition, Prentice Hall of India Learning,2010
2. Higher Engineering Mathematics, B.S.Grewal, 41ed, Khanna Publishers, 2012.
3. Complex Variables & Applications, R.V.Churchil, 6th Edition, Mc.Graw. Hill, 2009.
4. Engineering Mathematics-III, C. Shankaraiah, V.G.S Book Links, 2010.



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